

BIOMECHANICS (EPHE 341)

# Static Analysis

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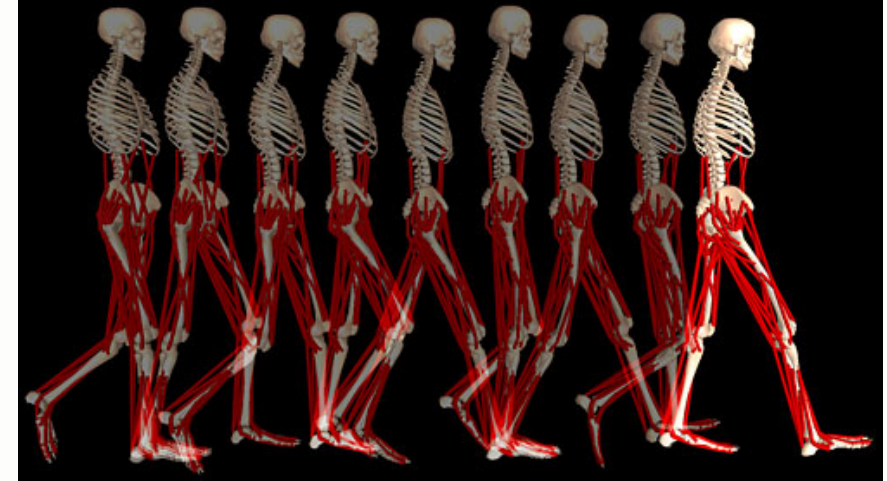
Two conditions, three equations, and the forces inside a body that nobody can measure.

## *Where this fits*

Through kinematics and kinetics we have been working toward being able to quantify human activity, both **in motion** and **at rest**.

With those concepts and the relationships Newton's laws give us, we can **solve for unknown variables** — which is the whole point, because the variables we most want are the ones we cannot measure.

You cannot put a load cell on a tendon in a living person. You can put a force plate under their foot and a camera in front of them. Static analysis is the bridge between those two facts.



Everything in red is a force nobody has ever directly measured in a walking human being.

## *Back to the first law*

The ability to solve for unknowns begins with Newton's laws.

**Linear.** An object at rest stays at rest, and an object in uniform motion stays in uniform motion, unless acted upon by a **net** external force.

**Angular.** A body maintains a constant state of angular motion unless acted on by a **net** external torque.

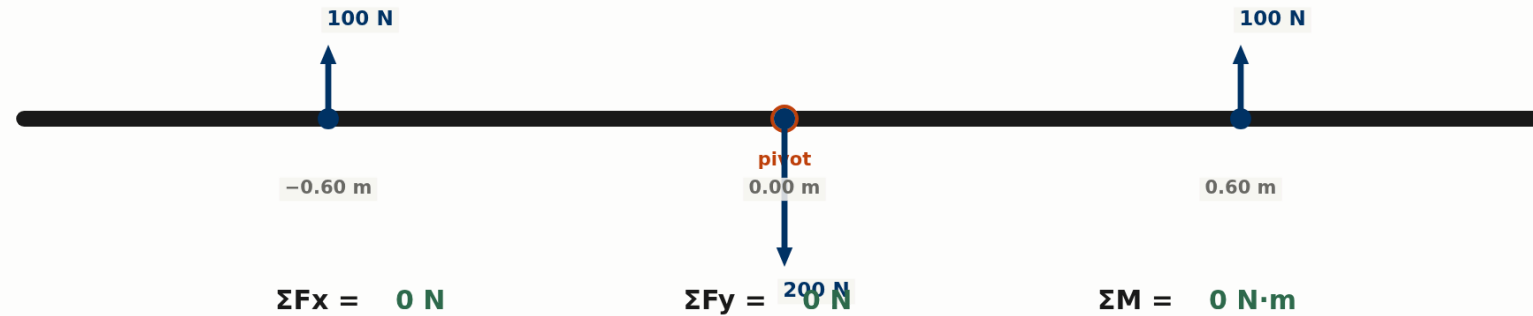
The operative word is **net**. We have to look at *all* the forces and *all* the moments acting on the system at once.

Which gives two cases, and only two:

- $\Sigma F$  and  $\Sigma M$  are **zero** → no acceleration. The system is in **equilibrium**.
- $\Sigma F$  or  $\Sigma M$  is **not zero** → the state of motion is changing. There is an acceleration.

Today is the first case. The next lecture is the second one, and the equations barely change — the right-hand sides stop being zero.

*Two conditions, and neither one implies the other*



**in static equilibrium — it will not move and it will not turn**

Drag a force along the bar, or press a case below · the bar is drawn moving whenever a sum is not zero · an object in static equilibrium satisfies BOTH conditions, and neither one implies the other

**In equilibrium**

Forces balance, moments do not

Moments balance, forces do not

Static equilibrium needs **two** conditions, and the second is the one people forget. Press the middle case: the two forces are equal and opposite, so  $\Sigma F$  is zero and the bar goes nowhere — and it spins, because they are not in line. Press the third: now the moments cancel and the forces do not, so it flies off without turning. Only the first case is equilibrium.

## *The equations of static equilibrium*

When the sum of all forces and all moments is zero, the object has zero acceleration and is said to be in **equilibrium**. These are the equations we use to solve for unknown quantities:

$$\Sigma F = 0 \quad \Sigma M = 0$$

They hold on every axis:

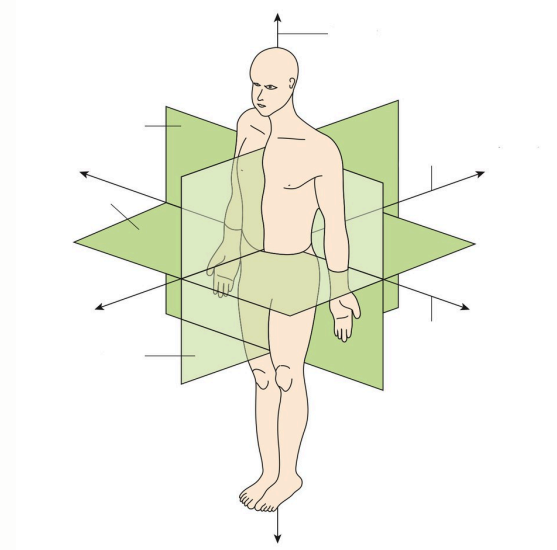
$$\Sigma F_x = 0 \quad \Sigma M_x = 0$$

$$\Sigma F_y = 0 \quad \Sigma M_y = 0$$

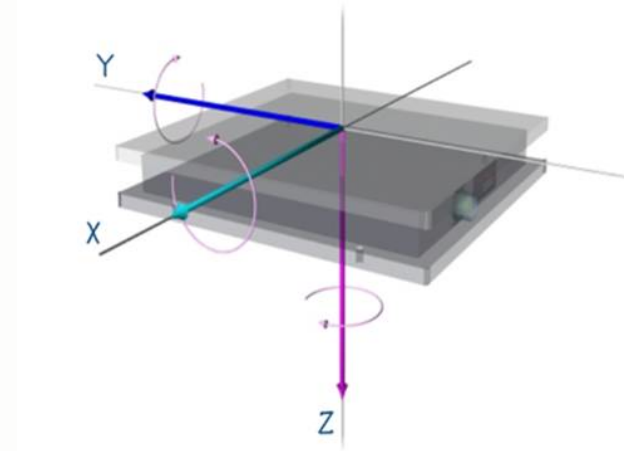
$$\Sigma F_z = 0 \quad \Sigma M_z = 0$$

**In this course we work in two dimensions,**  
which leaves three:

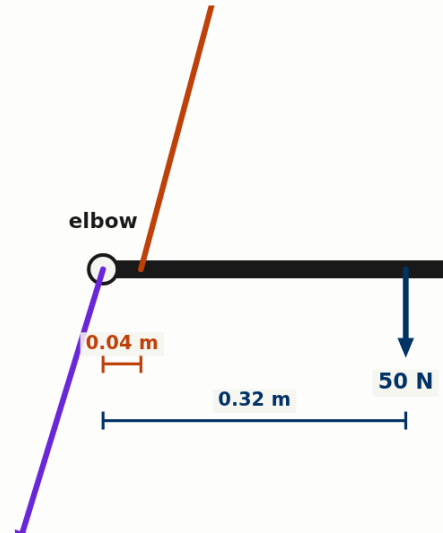
$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M = 0$$



↑ Figure 4.5: Axes of rotation and planes of movement of the human body



## The joint reaction force



1. the load makes a moment about the elbow

$$50 \times 0.32 = 16.0 \text{ N}\cdot\text{m}$$

2. the muscle must cancel it, from 0.04 m

$$M = 16.0 / (0.04 \sin 75^\circ) = 414 \text{ N}$$

3. the joint carries whatever is left over

$$R = 366 \text{ N, which is 7.3 times the load}$$

holding 50 N costs the elbow 366 N

The joint reaction is not the mirror of the load · it is the mirror of the load AND the muscle, and the muscle is always the larger of the two because its moment arm is always the shorter one

The load  50 N

How far out it is  0.32 m

Where the muscle inserts  0.040 m

Hold fifty newtons in your hand and your elbow carries several hundred. The reason is the third slider: the muscle inserts a few centimetres from the joint and the load is a third of a metre away, so the muscle has to be **many times larger** than the load just to balance its moment — and the joint has to carry both. Slide the insertion in toward the joint and watch what it costs.

## *Where the force acts matters — but only for the moment*

For the **force** equations, it does not matter where on the body a force acts. Forces just add:

$$\Sigma F = F_{\text{applied}} + F_{\text{joint reaction}} = 0$$

And in components, which is how we will actually use them:

$$\Sigma F_x = F_{ax} + F_{rx} = 0$$

$$\Sigma F_y = F_{ay} + F_{ry} = 0$$

For the **moment** equation it matters completely. The applied force tends to rotate the segment about the joint, and that moment has to be countered if the limb is to stay still:

$$\Sigma M = M_{\text{applied}} + M_{?} = 0$$

The question mark is the muscle, and solving for it is the entire exercise.

A force applied *at* the joint makes **no moment about that joint** — its moment arm is zero. That is why we take moments about the joint: the two unknown reactions vanish from the equation and the muscle force is left on its own.

# **The method**

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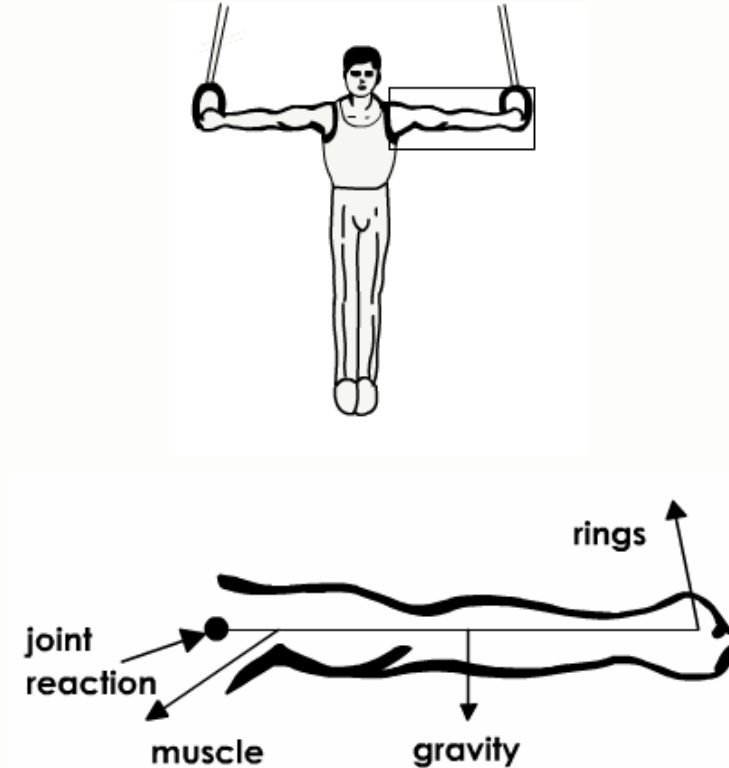
A free body diagram, a force table, three equations. Always in that order.

## Example problem — the iron cross

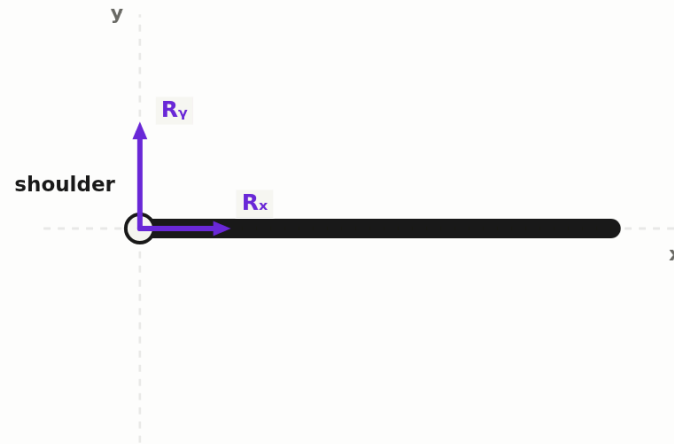
Using the diagram of the **right arm of the gymnast**, find the magnitude of the muscle force ( $M$ ) and the joint reaction forces at the shoulder.

### Given

- the arm is held horizontal
- force of the rings = **400 N**, located **0.88 m** from the shoulder
- the ring force acts at **80° above the negative X axis**
- arm mass = **3.4 kg**, located **0.39 m** from the shoulder
- the muscle force acts **0.08 m** from the shoulder
- the muscle force acts at **40° below the negative X axis**



## Step one: draw a free body diagram



Step 2 of 5

### Draw the reaction forces

Vertical ( $R_y$ ) and horizontal ( $R_x$ ) at the shoulder. You do not know their size yet, which is fine; you know they are there.

Every force on this diagram acts on the ARM, and nothing on it acts on anything else · that is what makes it a free body diagram, and it is why the three sums can be written down

Cut the arm free

**Draw the reaction forces**

Draw the muscle force

Draw the gravitational force

Draw the force of the rings

A free body diagram is not a drawing of the situation — it is a drawing of **one object with every force that acts on it and nothing else**. Step through his five. The one students skip is the first: cutting the arm away from the body, and replacing the body with two unknown forces. Until that cut is made there is nothing to sum.

## Step two: build a force table

To keep track of the equations and the variables in them, build a **force table**. One row per force; three columns for what that force contributes.

| Force | $F_x$  | $F_y$  | $M$ |
|-------|--------|--------|-----|
| F1    | $F1_x$ | $F1_y$ | M1  |
| F2    | $F2_x$ | $F2_y$ | M2  |
| $R_x$ | $R_x$  | 0      | 0   |
| $R_y$ | 0      | $R_y$  | 0   |
| Total | 0      | 0      | 0   |

The first column lists every force acting on the system. The three remaining columns are what each force contributes in  $x$ , in  $y$ , and as a moment.

Because of the laws of static equilibrium, **the sum of each column is zero**. So each column becomes an equation:

- column two  $\rightarrow \Sigma F_x = 0$
- column three  $\rightarrow \Sigma F_y = 0$
- column four  $\rightarrow \Sigma M = 0$

Look at the last two rows. The reactions act *at* the joint, so their moment arm is zero and their moment column entry is zero. **That is the whole trick**, written into the shape of the table.

His table, filling itself in

Force Table

| Force   | X component        | Y component       | Moment                  |
|---------|--------------------|-------------------|-------------------------|
| rings   | $-400\cos80^\circ$ | $400\sin80^\circ$ | $400(0.88\sin80^\circ)$ |
| gravity | 0                  | $-3.4(9.81)$      | $-3.4(9.81)(0.39)$      |
| muscle  | $-M\cos40^\circ$   | $-M\sin40^\circ$  | $-M(0.08\sin40^\circ)$  |
| $R_x$   | $R_x$              | 0                 | 0                       |
| $R_y$   | 0                  | $R_y$             | 0                       |

the three equations

$\Sigma M = 0$

$346.65 - 13.01 - M(0.0514) = 0$   
 $M = 6488 \text{ N}$

$\Sigma F_x = 0$

$-69.5 + 0 + -4970 + R_x = 0$   
 $R_x = 5040 \text{ N}$

$\Sigma F_y = 0$

$393.9 - 33.4 + -4171 + R_y = 0$   
 $R_y = 3810 \text{ N}$

his printed answers:  $M = 6488 \text{ N}$ ,  $R_x = 5040 \text{ N}$ ,  $R_y = 3810 \text{ N}$  — the same, to the newton

His slides 18 to 27 · 400 N at 0.88 m and 80°, a 3.4 kg arm at 0.39 m, the muscle at 0.08 m and 40° · every row is his, and the sliders move his givens

Ring force

400 N

Muscle angle

40°

Muscle insertion

0.080 m

His numbers

Build it up

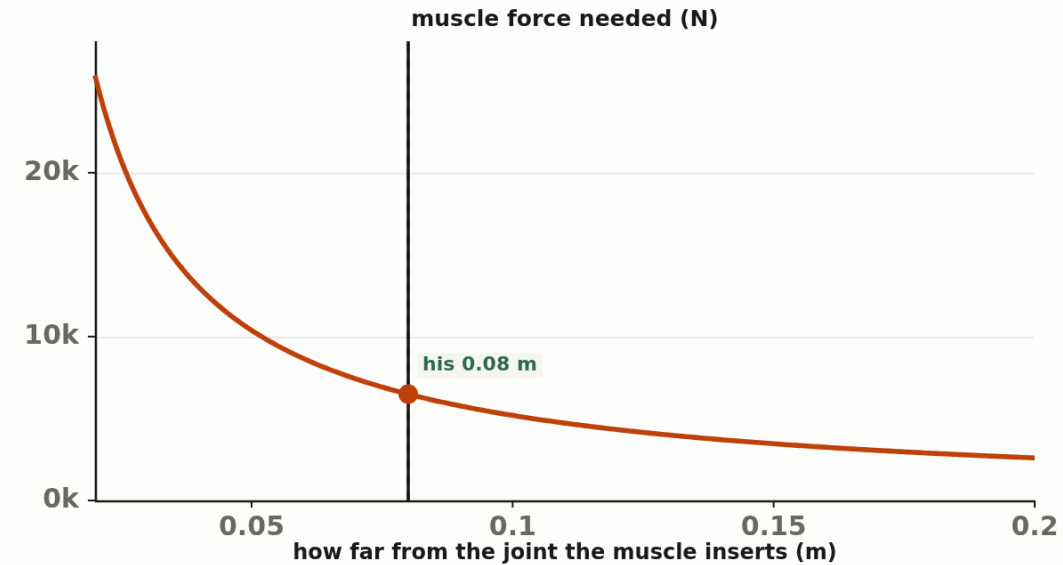
Show it all

▶ Fill the table

## Why the answer was so large



a 11.0 : 1 disadvantage



at 0.080 m the muscle needs 6488 N — 16.2 times the ring force, and 195 times the weight of the arm

Every other given is a choice; this one is anatomy · the curve is a hyperbola, so a centimetre of insertion distance is worth far more near the joint than far from it · that is also why a torn-off tendon reattached a centimetre too close is a permanent loss of strength

Where the  
muscle inserts

0.080 m

The answer was 6488 N because of one ratio: the rings pull from 0.88 m and the muscle pulls from 0.08 m. **Muscles are always on the wrong end of the lever**, and they are so for a reason — a short moment arm means a small muscle shortening produces a large, fast movement at the hand. You pay for speed and range with force.

## *The three equations, written out*

$\Sigma M = 0$  about the shoulder — the reactions have no moment there, so this equation has one unknown:

$$400(0.88 \sin 80^\circ) - 3.4(9.81)(0.39) - M(0.08 \sin 40^\circ) = 0$$

$$346.65 - 13.01 = M(0.05142)$$

$$M = \mathbf{6488 \text{ N}}$$

$\Sigma F_x = 0$ , now that  $M$  is known:

$$-400 \cos 80^\circ - 6488 \cos 40^\circ + R_x = 0$$

$$R_x = \mathbf{5040 \text{ N}}$$

$\Sigma F_y = 0$ :

$$400 \sin 80^\circ - 3.4(9.81) - 6488 \sin 40^\circ + R_y = 0$$

$$R_y = \mathbf{3810 \text{ N}}$$

Three equations, solved **in sequence** and never simultaneously, because the first one was arranged to have only one unknown in it.

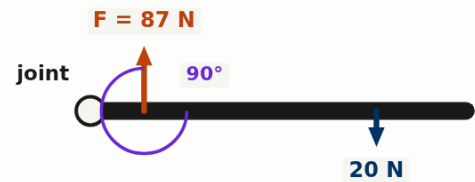
Put the answers in human terms before moving on.

**6488 N** of muscle force is about **195 times the weight of the arm** it is holding up, and **16 times** what the rings are pulling with.

The shoulder joint is carrying a resultant of about **6300 N** — roughly nine times the gymnast's whole body weight, through one shoulder, while he holds perfectly still.

Two rings at 400 N support 800 N in total, which is an 81 kg gymnast — so the given is self-consistent, and worth checking that way.

## The same method, three more times



Forearm, everything horizontal

moments about the joint

$$\Sigma M = 0$$

the load

$$20 \text{ N} \times 0.13 \text{ m} = 2.60 \text{ N}\cdot\text{m}$$

the muscle

$$F \times 0.03 \text{ m}$$

so

$$F = 2.60 / 0.03 = 86.7 \text{ N}$$

**86.7 N**      the Tutor's answer too

The muscle pulls 4.3 times harder than the weight it is holding, because its moment arm is 4.3 times shorter.

The Biomechanics Tutor, Static Equilibrium · moments about the joint first, so the two unknown reactions have no moment and drop out · only then the two force sums

Forearm

Shoulder

Tilted forearm

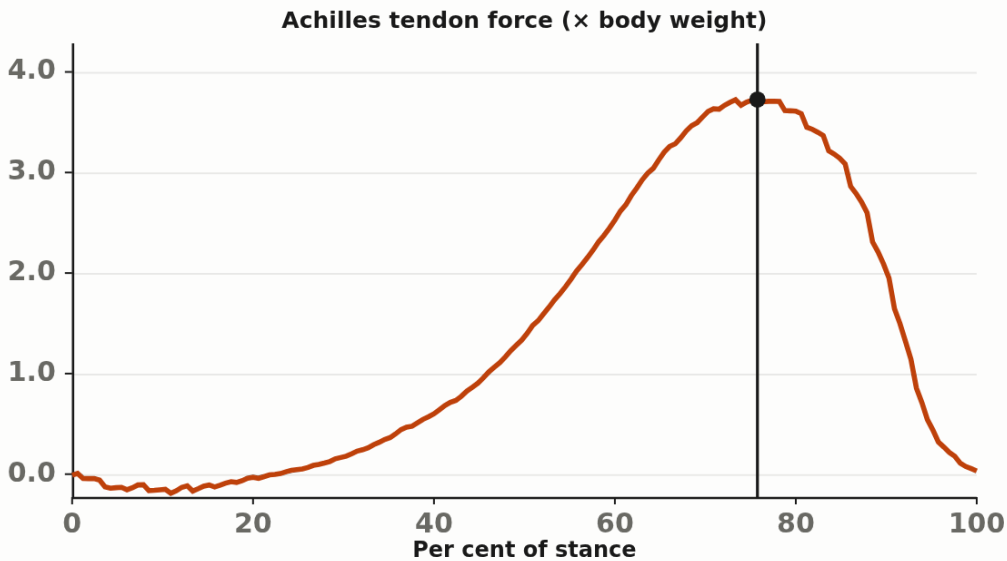
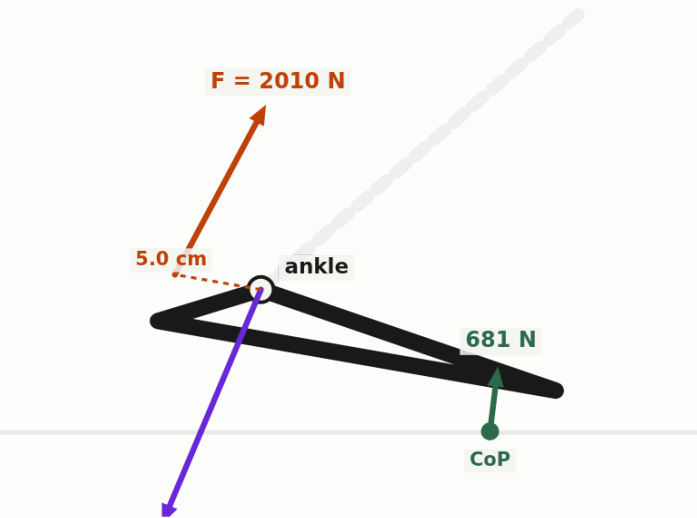
Three problems from the Tutor, in the order of difficulty you will meet them. The method never changes: free body diagram, force table, **moments about the joint first** because the reactions have no moment there, then the two force sums for the reactions. What changes is only how hard the moment arm is to read, and the third case is where you have to stop reading it and start computing it.

## **Does it work on real people?**

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His slide 28 says these forces cannot be measured. So let us compute some and see how close they land.

The same three equations, on a measured walking stance



at 76% of stance the floor pushes with  $1.26 \times$  body weight, the tendon pulls with 3.73, and the joint carries 4.92

Measured · Subject16 · 55 kg · plate at 240 Hz and measured markers · the one number here that is NOT measured is the tendon's moment arm, which the literature puts at 4.7 to 5.3 cm — it is the slider, so you can see what it is worth

Through  
stance

76%

Tendon  
moment arm

5.0 cm

Tendon force

Joint reaction

Check against inverse dynamics

► Roll through stance

The gymnast again, on someone walking. The foot is the free body; the ground force and where it acts are measured, the ankle is measured, and moments about the ankle give the tendon force. At the peak the Achilles is pulling with **3.7 times body weight** and the ankle joint is carrying **4.9** — while the floor is

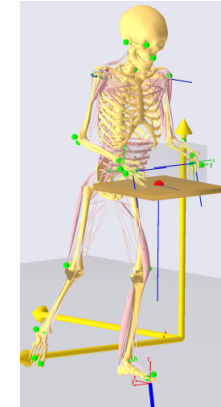
## *Why static analysis matters*

It is practically impossible to measure internal forces directly. The ability to solve static equilibrium problems **yields an estimate of them** — and for most of what biomechanics knows about the inside of a living body, an estimate like this is all there has ever been.

In the gymnast, the muscle force needed to hold that position was extremely large, and that in turn means a very large stress inside the shoulder joint.

## **Where it gets used**

- **Ergonomics** — how much spinal compression a lift costs, and where to put the limit
- **Static postures** — holding, carrying, standing at a bench
- **Sport** — any held position, and the instant of peak load in a moving one



The honest caveat, from the last figure: a static analysis ignores the segment's own weight and acceleration. On a real walking stance that made it about **15% high**. It is an estimate, and knowing the size of its error is part of using it.

## *Signs, and the two mistakes everybody makes*

**1. Components.** Resolve every force the same way, from the same axes, every time:

$$F_x = F \cos\theta \quad F_y = F \sin\theta$$

with  $\theta$  measured **anticlockwise from the positive x axis**, always.

His givens are phrased relative to the *negative* x axis, so convert them once, at the start, and write the converted number down:

- $80^\circ$  above the negative x axis  $\rightarrow \theta = 100^\circ$
- $40^\circ$  below the negative x axis  $\rightarrow \theta = 220^\circ$

Then cos and sin give you the signs for free, and you never have to decide whether something is “negative” by looking at the picture.

**2. Moments.** Pick a sense — anticlockwise positive is standard — and keep it for the whole problem:

$$M = F d_\perp = F d \sin\theta$$

where  $\theta$  is the angle **between the force and the segment** — not between the force and the horizontal.

When the segment is horizontal,  $d \sin\theta$  is just the vertical component times the distance, which is why the horizontal cases feel easy.

When it is not, that shortcut silently fails. The Tutor’s tilted-forearm problem is in the deck for exactly this reason.

## What to take away

### The two conditions

- $\Sigma F = 0$  stops it going anywhere;  $\Sigma M = 0$  stops it turning.  
**Neither implies the other.**
- In 2D that is three equations, which solves three unknowns — and biomechanics problems are built with exactly three.

### The method, in order

1. Cut the segment free and draw every force on it — the free body diagram.
2. Build the force table: a row per force, three columns.
3. **Moments about the joint first**, because the reactions have no moment there and drop out, leaving one unknown.
4. Then  $\Sigma F_x$  and  $\Sigma F_y$ , one reaction each.

### What it tells you

- Muscles have short moment arms and loads have long ones, so **muscle forces are always large multiples of the load** — 16× in the gymnast.
- The joint carries the load *and* the muscle, so **joint forces are larger still** — 9× body weight through one shoulder, 4.9× through an ankle in ordinary walking.
- None of those numbers can be measured. All of them can be computed, from a force plate and a camera.

And the honest limit: a static analysis ignores the segment's own weight and acceleration. Measured against a full inverse-dynamics solution of the same stride, it ran about **15% high**. **Next lecture** the zeros become *ma*, and that error goes away.